

# EECS 4413 Group Project

## Deliverable 1

### Purpose of the First Deliverable

The purpose of this deliverable is to produce:

- a) A collection of sequence diagrams for 4 use cases presented in the Project Description Document
- b) A collection of activity diagrams for 4 use cases presented in the Project Description Document
- c) A detailed architecture specification of your system as a UML component diagram
- d) An implementation plan in the form of a GANTT Diagram  
([https://en.wikipedia.org/wiki/Gantt\\_chart](https://en.wikipedia.org/wiki/Gantt_chart))

In this Assignment, you will learn to:

- Specify the use cases of an e-commerce system using sequence and activity diagrams
- Decompose a complex problem into sub-problems
- Design and specify the architecture of an e-commerce system
- Produce a simple Software Design Document (SDD) for an e-commerce system from use cases

### Due

Please check e-class regularly for any updates regarding the submission process and deadlines. One submission per group.

### Late Penalty

Late project will be accepted for up to two days after the due date. Submissions late up to 24 hours will incur a penalty of 20%. No submissions after 48<sup>th</sup> hour of the deadline are accepted. The same time windows hold if there is an extension is given by the instructor.

### Group Effort

This stage of the project is expected to be a group effort, with each member of the group contributing equally in a reasonable fashion. If it is determined that you are guilty of cheating on the assignment, you could receive a grade of zero with a notice of this offence submitted to the Dean of your home faculty for inclusion in your academic record.

### What to Hand in

Your submission, as noted above, will be electronically in the form of an archive (e.g. zip) file. Please check e-class regularly for any changes related to the submission process. You are to complete the deliverable by providing the required documentation using the template provided to you in the Project Resources section and

# Assignment Task

In this deliverable, you will craft a simple Software Design Document (SDD) that captures key ideas of the software design of your system that meets the requirements of the use cases presented in the Project Description Document.

Your SDD documentation template must consist of the following sections and content:

## 1. Main Page

The main page will contain:

- A title for the project, and a sub-title indicating that this is the requirements documentation for that project.
- The Document Change Control revision history of your document in tabular form. It should be updated as people add to and edit your requirements documentation. It should be formatted as follows:

| Version | Date | Author(s) | Summary of Changes |
|---------|------|-----------|--------------------|
|         |      |           |                    |
|         |      |           |                    |

- The table of Contents.

## 2. Introduction

The Introduction section, should have the following content.

**Purpose:** This sub-section outlines the general objectives of the project.

**Overview:** This sub-section provides an overview of the problem being solved by this software and the requirements of the system. This should be an executive summary.

**References:** A list of references to other documents that might provide context or otherwise assist in the understanding of this document.

## 3. Major Design Decisions

Text describing significant design choices, and modularization criteria. Modularization criteria include high cohesion and low coupling. Explain how your design aims to achieve these criteria.

Also, describe what architectural patterns have you chosen for your application. Describe at least one architectural pattern that you have considered but have not chosen.

## 4. Sequence Diagrams

Sequence diagrams for 4 use cases described in the Project Description document. Make sure that in each diagram, you use correct and complete notation capturing the corresponding use case.

Please make diagrams self-explanatory (e.g. use correct naming conventions, parameters, returns etc.)

## 5. Activity Diagrams

Activity diagrams for 4 use cases described in the Project Description document. Use different use case than those you used in Sequence Diagrams. Make sure that in each diagram, you use correct and complete notation capturing the corresponding use case. Please make diagrams self-explanatory (e.g. use correct naming conventions, parameters, data generated and consumed etc.)

## 6. Architecture

Provide an initial decomposition of your system in terms of services and interactions. At this stage you can use a block and arrow diagram or a UML component diagram.

Provide a description of each service and its interaction with other services in the form of the following tables. You only need to describe the Façade of the service ( check the design pattern façade). You can use either Java or REST interfaces.

| Services     |                       |                         |                             |
|--------------|-----------------------|-------------------------|-----------------------------|
| Service Name | Description           | Exposed Interface Names | Interface Description       |
| M1           | “a description”       | M1:I1                   | M1:I1 “a description”       |
|              |                       | M1:I2                   | M1:I2 “another description” |
| M2           | “another description” | M2:I3                   | M2:I3 “a third description” |

| Interfaces     |                                               |                              |
|----------------|-----------------------------------------------|------------------------------|
| Interface Name | Operations                                    | Operation Descriptions       |
| M1:I1          | <return type> I1:Op1() used by M2, M4         | M1:I1:Op1(): “a description” |
|                | <return type> I1:Op2(int x) used by M3 and M9 | M1:I1:Op2(int x): “xxxxx”    |
| M1:I2          | <return type> M1:I2:Op3() used by M9          | M1:I2:Op3() used by M1       |

Here we have two Services M1, and M2. M1 exposes two interfaces I1, and I2. Interface I1 has two operations Op1(), and Op2(int x), while I2 has one operation Op3(). Please note that the return type is included in the Operations column. This section will be revised in deliverables 2 & 3 to depict a complete view of your final system.

## 7. Activities Plan, Product Backlog, and Sprint Backlog

In this section provide the following:

1. a complete product backlog list (i.e. what needs to be implemented as a whole). Identify the list of items to be implemented for each deliverable out of the backlog list. You are expected every four weeks to implement, refactor, integrate, and run your code.
2. A GANTT diagram with a schedule of your planned activities

This section of your report may be slightly revised in deliverable 2.

### Group Meeting Logs

In this Section you write minutes of each meeting, list the meeting attendance, what the topics of discussion in the meeting were, any decisions that were made, and which team members were assigned which tasks. These minutes must be submitted with the project report in each deliverable and will provide input to be used for the overall assessment of the project.

## 8. Test Driven Development (TDD)

At this point you do not have fully implemented your code yet, but hopefully you have some use cases implemented.

In this section, you will provide an initial set of test cases you consider fundamental for verifying your requirements if you had assumed you have the code implemented. Following then the concept of TDD, you write your code so that these tests will pass. Of course, testing assumes that more tests should be written once the code is implemented, but this is a starting point we can use for our course here. You should provide a more comprehensive list of test cases with your Deliverables 2 and 3. For this deliverable provide a list of at least 16 test cases (4 per group member) in the form of a table as follows:

|                       |                                                                                                                                                                                                                                                                                                                      |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Test ID               | The unique Id of the test case                                                                                                                                                                                                                                                                                       |
| Category              | Which part of the system is tested ( <i>e.g. evaluation of user credentials stored on file or DB</i> )                                                                                                                                                                                                               |
| Requirements Coverage | The unique ID of the requirement tested ( <i>e.g. UC1-Successful-User-Login</i> )                                                                                                                                                                                                                                    |
| Initial Condition     | Initial conditions required for the test case to run ( <i>e.g. the system has been initiated and runs</i> )                                                                                                                                                                                                          |
| Procedure             | The list of steps required for this test case ( <i>e.g.</i><br><ol style="list-style-type: none"> <li>1. The user selects login</li> <li>2. The user provides a user name</li> <li>3. The user provides a password</li> <li>4. The user logs-in into the system and is presented with the main UI window)</li> </ol> |
| Expected Outcome      | The expected outcome of the test case ( <i>e.g. the login form closes, and the user is presented with the main UI window</i> )                                                                                                                                                                                       |
| Notes                 | Any other notes you may want to add for this test case, which are also reflected in the requirements specification ( <i>e.g. the user should provide only alphanumeric user names and passwords without any special characters</i> )                                                                                 |

From the point you complete and submit this deliverable you will keep on working on your implementation and complete as required the list of any additional test cases you want to include.

That is this section of your report will be revised in deliverables 2 & 3 to depict a complete picture by the end of the project.

## **8. Acknowledgment**

Here you document the use of AI. You can use AI to increase your productivity. However, you are required to understand, explain and critique, change and maintain any artifact generated with the help of AI.